

Chapter-4 Techniques of Circuit Analysis

Node: A point where two or more circuit elements join.

Essential Node: A node where three or more circuit elements join.

Path: A trace of adjoining basic elements with no elements included more than once.

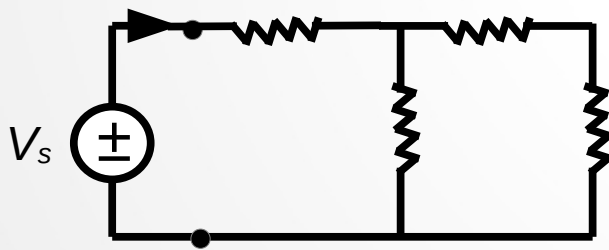
Branch: A path that connects two nodes.

Essential Branch: A path which connects two essential nodes without passing through an essential node.

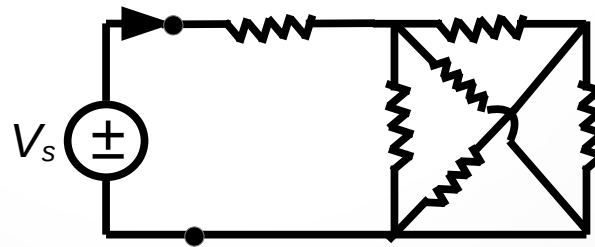
Loop: A path whose ending node is the same as the starting node.

Mesh: A loop that does not enclose any other loop.

Planer circuit: A circuit that can be drawn on a plane with no crossing branches.



Planer circuit



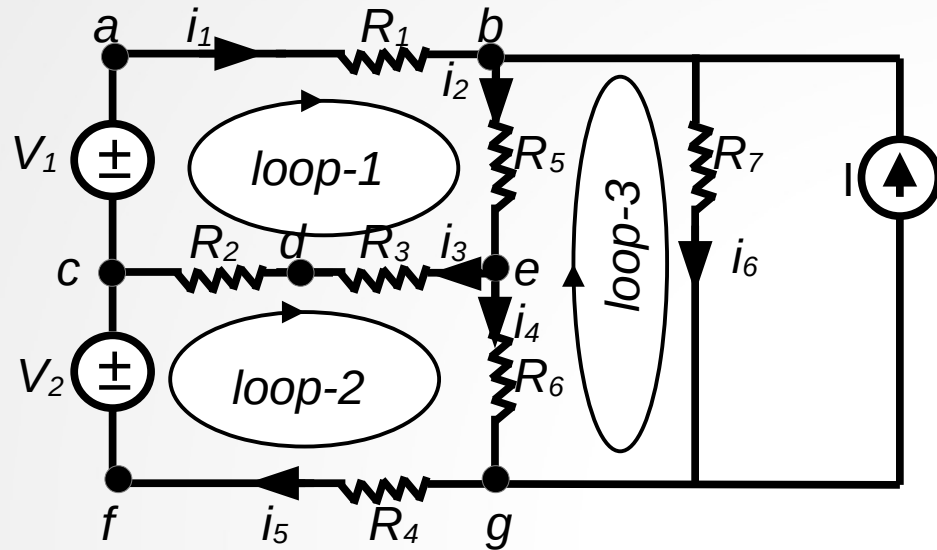
Not a planer circuit

Chapter-4 Techniques of Circuit Analysis

- How many equations do we need?
- The number of unknown currents in a circuit equals to the number of branches, b , where current is not known.
- n : number of nodes.
- We can derive $n-1$ independent equations by applying KCL. Need $b-(n-1)$ more equations. We can write $b-(n-1)$ equations using KVL.
- In fact the number of unknown currents in a circuit equals to the number of essential branches, b_e , where current is not known.
- We can derive n_e-1 independent equations by applying KCL. Need $b_e-(n_e-1)$ more equations. We can write $b_e-(n_e-1)$ equations using KVL.
- n_e is the number essential nodes.

Chapter-4 Techniques of Circuit Analysis

Example: The systematic Approach--an Illustration



There are four essential nodes. We can write three independent equations using KCL

$$\text{KCL for node b: } -i_1 + i_2 + i_6 - I = 0$$

$$\text{KCL for node c: } i_1 - i_3 - i_5 = 0$$

$$\text{KCL for node e: } i_3 + i_4 - i_2 = 0$$

There are six unknowns. We need three more independent equations. There are four loops. We can write three more independent equations using KVL for these loops.

$$\text{KVL for loop-1: } R_1 * i_1 + R_5 * i_2 + (R_2 + R_3) * i_3 - V_1 = 0$$

$$\text{KVL for loop-2: } -(R_2 + R_3) * i_3 + R_6 * i_4 + R_4 * i_5 - V_2 = 0$$

$$\text{KVL for loop-3: } -R_5 * i_2 + R_7 * i_6 - R_6 * i_4 = 0$$

After rearranging, we get:

$$-i_1 + i_2 + 0 * i_3 + 0 * i_4 + 0 * i_5 + i_6 = I$$

$$i_1 + 0 * i_2 - i_3 + 0 * i_4 - i_5 + 0 * i_6 = 0$$

$$0 * i_1 - i_2 + i_3 + i_4 + 0 * i_5 + 0 * i_6 = 0$$

$$R_1 * i_1 + R_5 * i_2 + (R_2 + R_3) * i_3 + 0 * i_4 + 0 * i_5 + 0 * i_6 = V_1$$

$$0 * i_1 + 0 * i_2 - (R_2 + R_3) * i_3 + R_6 * i_4 + R_4 * i_5 + 0 * i_6 = V_2$$

$$0 * i_1 - R_5 * i_2 + 0 * i_3 - R_6 * i_4 + 0 * i_5 + R_7 * i_6 = 0$$

$$\Rightarrow \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 & -1 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 \\ R_1 & R_5 & R_2 + R_3 & 0 & 0 & 0 \\ 0 & 0 & -(R_2 + R_3) & R_6 & R_4 & 0 \\ 0 & -R_5 & 0 & -R_6 & 0 & R_7 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \\ i_6 \end{bmatrix} = \begin{bmatrix} I \\ 0 \\ 0 \\ V_1 \\ V_2 \\ 0 \end{bmatrix}$$

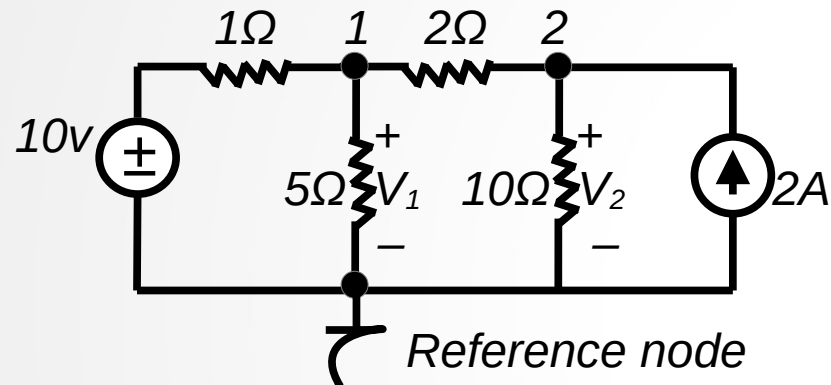
$$Ax = b \rightarrow x = A^{-1}b$$

Chapter-4 Techniques of Circuit Analysis

Introduction to the Node-Voltage Method

- This method enable us to describe a circuit in terms of n_e-1 equations.
- A node voltage is defined as the voltage rise from the reference node to a nonreference node.

Example:



V_1 is the node voltage for node-1

V_2 is the node voltage for node-2

KCL for node-1

$$\frac{V_1 - 10}{1} + \frac{V_1}{5} + \frac{V_1 - V_2}{2} = 0$$

KCL for node-2

$$\frac{V_2 - V_1}{2} + \frac{V_2}{10} - 2 = 0$$

$$\Rightarrow 17V_1 - 5V_2 = 100$$

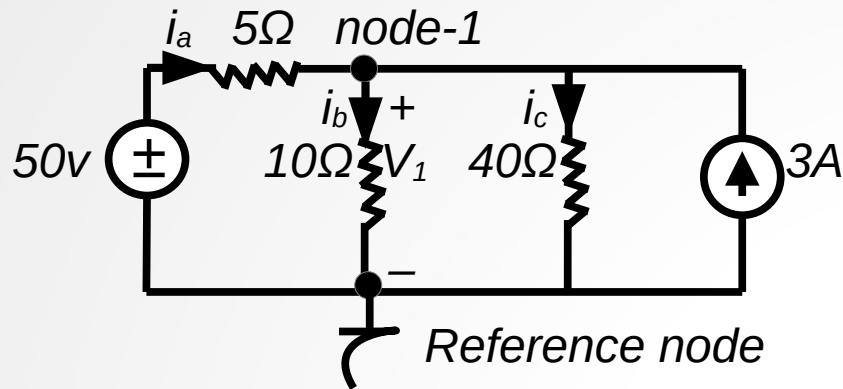
$$6V_2 - 5V_1 = 20$$

$$\Rightarrow 77V_1 = 700 \Rightarrow V_1 = \frac{100}{11}$$

$$\Rightarrow V_2 = \frac{5V_1 + 20}{6} = \frac{120}{11}$$

Chapter-4 Techniques of Circuit Analysis

Example: Find i_a, i_b, i_c and power for each circuit elements



V_1 is the node voltage for node-1

KCL for node-1

$$\frac{V_1 - 50}{5} + \frac{V_1}{10} + \frac{V_1}{40} - 3 = 0$$

$$\Rightarrow V_1 = 40 \text{ V}$$

$$\Rightarrow i_a = \frac{50 - V_1}{5} = \frac{50 - 40}{5} = 2 \text{ A}$$

$$\Rightarrow i_b = \frac{40 - 0}{10} = 4 \text{ A}$$

$$\Rightarrow i_c = \frac{40 - 0}{40} = 1 \text{ A}$$

Powers for the circuit elements:

$$P_{50\text{V}} = -50 * i_a = -100 \text{ W (generates)}$$

$$P_{3\text{A}} = -3 * V_1 = -3 * 40 = -120 \text{ W (generates)}$$

$$P_{10\Omega} = i_b^2 * 10 = 4^2 * 10 = 160 \text{ W (consumes)}$$

$$P_{5\Omega} = i_a^2 * 5 = 2^2 * 5 = 20 \text{ W (consumes)}$$

$$P_{40\Omega} = i_c^2 * 40 = 1^2 * 40 = 40 \text{ W (consumes)}$$

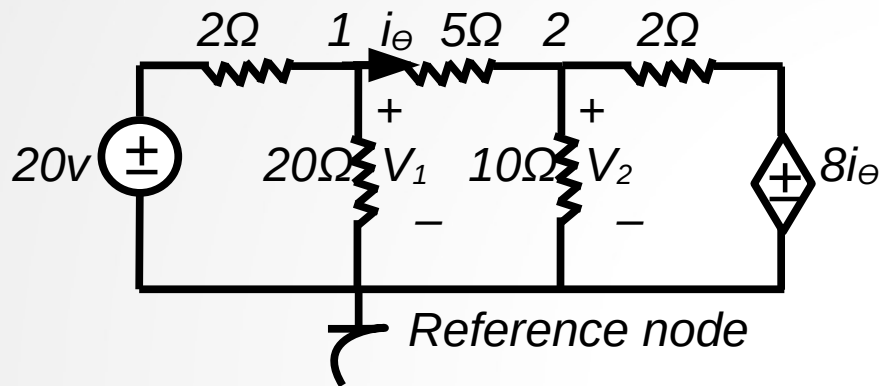
$\Rightarrow$ Generated power = Consumed power

Chapter-4 Techniques of Circuit Analysis

Node-Voltage Method and Dependent Sources

-Use the equations imposed by the dependent sources.

Example: $P_{5\Omega} = ?$



There are three essential nodes.

We can write two node-voltage equations

V_1 is the node voltage for node-1

V_2 is the node voltage for node-2

Constraint equation (i_θ)

$$i_\theta = \frac{V_1 - V_2}{5}$$

KCL for node-1

$$\frac{V_1}{20} + \frac{V_1 - 20}{2} + \frac{V_1 - V_2}{5} = 0$$

$$\Rightarrow 15V_1 - 4V_2 = 200$$

KCL for node-2

$$\frac{V_2}{10} + \frac{V_2 - 8i_\theta}{2} = i_\theta \quad \text{or} \quad \left(\frac{V_2}{10} + \frac{V_2 - 8i_\theta}{2} + \frac{V_2 - V_1}{5} = 0 \right)$$

$$\Rightarrow V_2 + 5V_2 - 40i_\theta = 10i_\theta \Rightarrow V_2 + 5V_2 - 50i_\theta = 0$$

$$\Rightarrow 6V_2 - 50 \frac{V_1 - V_2}{5} = 0 \Rightarrow 16V_2 = 10V_1$$

$$\Rightarrow 24V_2 - 4V_2 = 200 \Rightarrow V_2 = 10\text{V}$$

$$\Rightarrow V_1 = 1.6V_2 = 16\text{V}$$

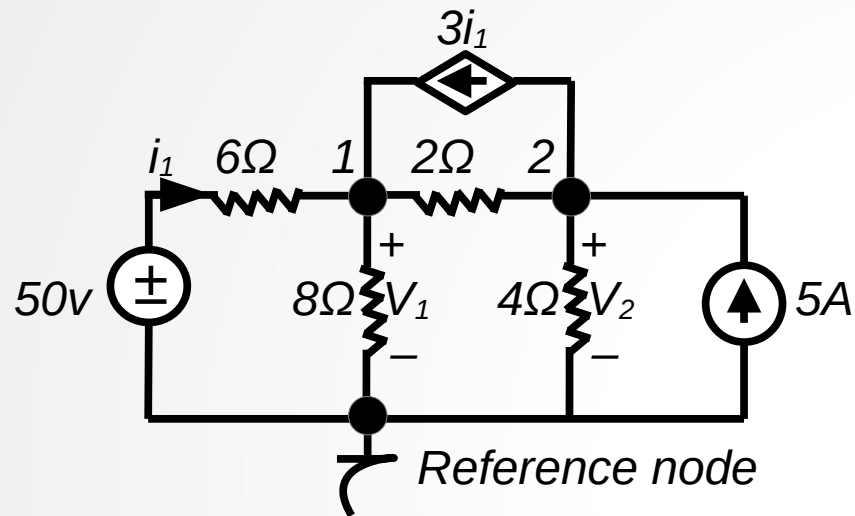
$$i_\theta = \frac{V_1 - V_2}{5} = \frac{16 - 10}{5} = 1.2\text{A}$$

$$P_{5\Omega} = i_\theta^2 * 5 = 7.2\text{W}$$

Chapter-4 Techniques of Circuit Analysis

Example:

- Use the node-voltage method to find the power associated with each source in the circuit
- State whether the source is delivering power to the circuit or extracting power from the circuit.



Constraint equation: $i_1 = \frac{50 - V_1}{6}$

KCL for node-1

$$\begin{aligned} \frac{V_1}{8} + \frac{V_1 - 50}{6} - 3i_1 + \frac{V_1 - V_2}{2} &= 0 \\ \Rightarrow \frac{V_1}{8} + 4 \frac{V_1 - 50}{6} + \frac{V_1 - V_2}{2} &= 0 \\ \Rightarrow 31V_1 - 12V_2 &= 800 \end{aligned}$$

KCL for node-2

$$\begin{aligned} \frac{V_2}{4} - 5 + 3i_1 + \frac{V_2 - V_1}{2} &= 0 \\ \Rightarrow \frac{V_2}{4} - 5 + \frac{50 - V_1}{2} + \frac{V_2 - V_1}{2} &= 0 \\ \Rightarrow 3V_2 - 4V_1 &= -80 \\ \Rightarrow V_1 = 32\text{ V}, \quad V_2 = 16\text{ V} \\ \Rightarrow i_1 = \frac{50 - V_1}{6} = \frac{18}{6} &= 3\text{ A} \end{aligned}$$

$P_{5A} = -V_2 * 5 = -80\text{ W}$ Delivers power

$P_{3i_1} = (V_2 - V_1) * 3i_1 = (16 - 32) * 3 * 3$
 $= -144\text{ W}$ Delivers power

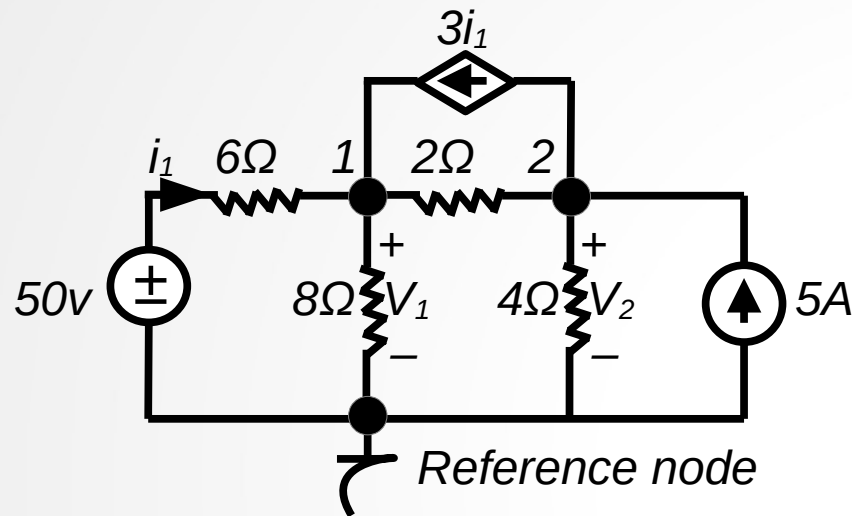
$P_{50V} = -50 * i_1 = -50 * 3 = -150\text{ W}$ Delivers power

All sources are delivering power to the circuit

Chapter-4 Techniques of Circuit Analysis

Example:

- Use the node-voltage method to find the power associated with each source in the circuit
- State whether the source is delivering power to the circuit or extracting power from the circuit.



Constraint equation: $i_1 = \frac{50 - V_1}{6}$

KCL for node-1

$$\begin{aligned} \frac{V_1}{8} + \frac{V_1 - 50}{6} - 3i_1 + \frac{V_1 - V_2}{2} &= 0 \\ \Rightarrow \frac{V_1}{8} + 4\frac{V_1 - 50}{6} + \frac{V_1 - V_2}{2} &= 0 \\ \Rightarrow 31V_1 - 12V_2 &= 800 \end{aligned}$$

KCL for node-2

$$\begin{aligned} \frac{V_2}{4} - 5 + 3i_1 + \frac{V_2 - V_1}{2} &= 0 \\ \Rightarrow \frac{V_2}{4} - 5 + \frac{50 - V_1}{2} + \frac{V_2 - V_1}{2} &= 0 \\ \Rightarrow 3V_2 - 4V_1 &= -80 \\ \Rightarrow V_1 = 32\text{ V}, \quad V_2 = 16\text{ V} \\ \Rightarrow i_1 = \frac{50 - V_1}{6} = \frac{18}{6} &= 3\text{ A} \end{aligned}$$

$P_{5A} = -V_2 * 5 = -80\text{ W}$ Delivers power

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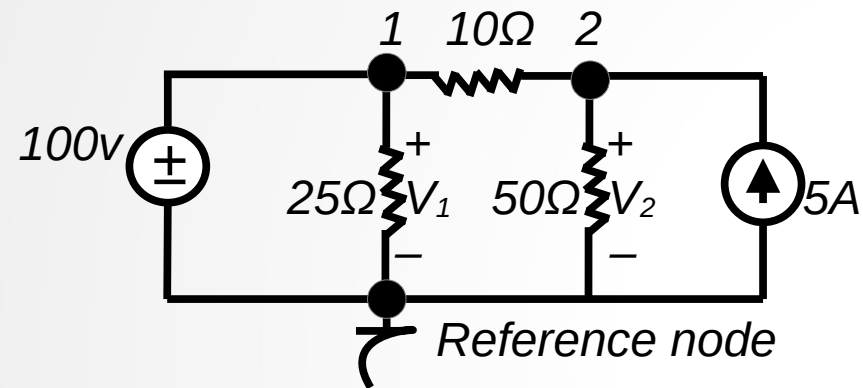
All sources are delivering power to the circuit

Chapter-4 Techniques of Circuit Analysis

Node-Voltage Method: Some Special Cases

-When a voltage source is the only element between two essential nodes, the node-voltage method is simplified.

Example



There are three essential nodes. V_1 is known.

There is only one unknown. Need one equation.

$$V_1 = 100 \text{ V}$$

KCL for node-2

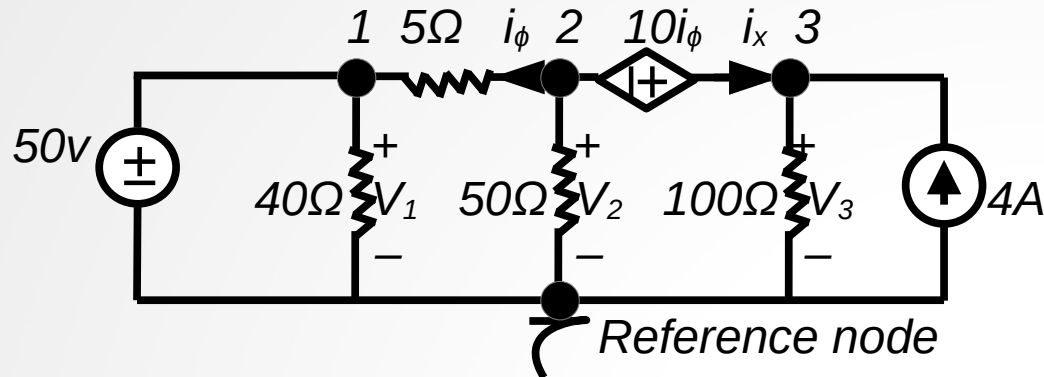
$$\frac{V_2 - V_1}{10} + \frac{V_2}{50} - 5 = 0$$

$$\Rightarrow 6V_2 - 5V_1 = 250 \Rightarrow 6V_2 = 750 \Rightarrow V_2 = 150 \text{ V}$$

In general, when you use the node-voltage method to solve circuits that have voltage sources connected directly between essential nodes, the number of unknown node voltage is reduced.

Chapter-4 Techniques of Circuit Analysis

Example: Find the node voltages



We do not need equation for node-1

We already know $V_1 = 50\text{ v}$

Constraint equation $i_\phi = \frac{V_2 - 50}{5}$

Constraint equation $V_3 = V_2 + 10i_\phi = 3V_2 - 100$

KCL for node-2

$$\frac{V_2 - V_1}{5} + \frac{V_2}{50} + i_x = 0$$

KCL for node-3

$$-i_x + \frac{V_3}{100} - 4 = 0$$

$$\Rightarrow \frac{V_2 - V_1}{5} + \frac{V_2}{50} + \frac{V_3}{100} - 4 = 0 \text{ -----equation-1}$$

$$\Rightarrow \frac{V_2 - 50}{5} + \frac{V_2}{50} + \frac{3V_2 - 100}{100} - 4 = 0$$

$$\Rightarrow 20V_2 + 2V_2 + 3V_2 - 1000 - 100 - 400 = 0$$

$$\Rightarrow V_2 = \frac{1500}{25} = 60\text{ v}$$

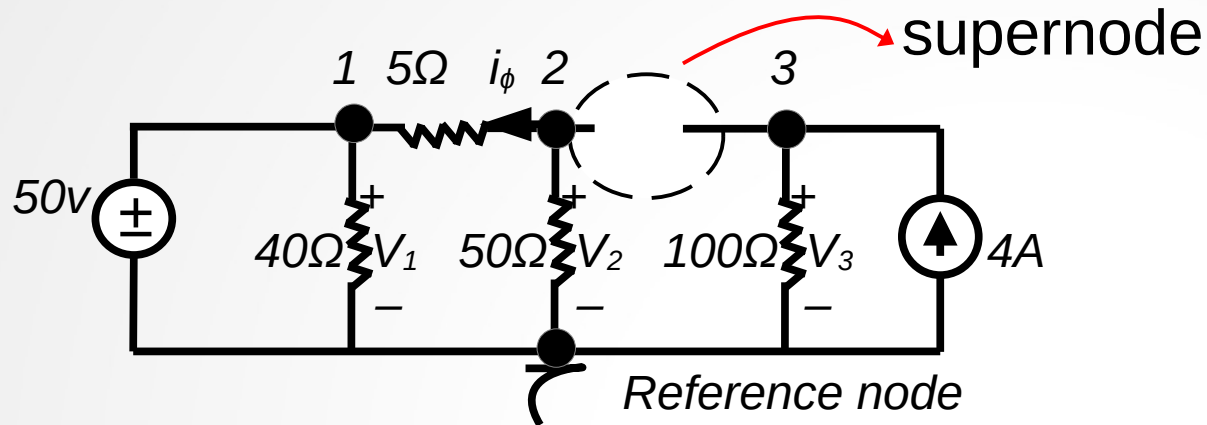
$$\Rightarrow i_\phi = \frac{V_2 - 50}{5} = \frac{10}{5} = 2\text{ A}$$

$$\Rightarrow V_3 = V_2 + 10i_\phi = 60 + 20 = 80\text{ v}$$

Chapter-4 Techniques of Circuit Analysis

The concept of a supernode

Equation-1 can be directly written



We can directly write equation-1 by combining node-1 and node-2

$$\frac{V_2 - V_1}{5} + \frac{V_2}{50} + \frac{V_3}{100} - 4 = 0 \text{ ----- equation-1}$$

$$V_1 = 50 \text{ v}$$

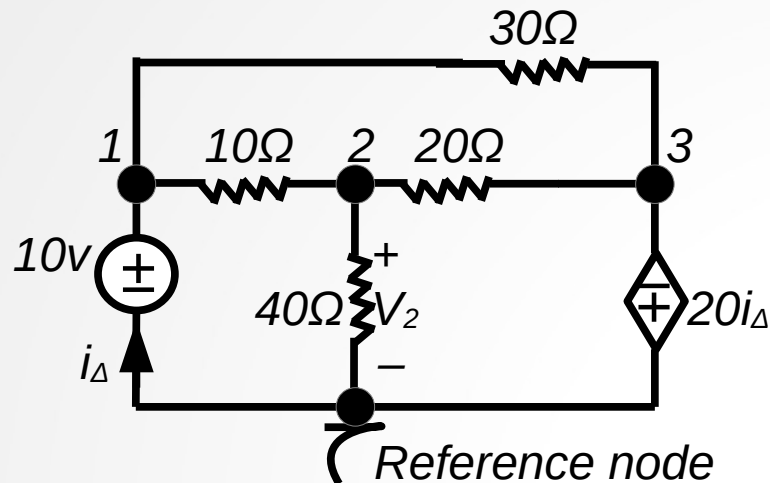
$$\text{Constraint equation } i_\phi = \frac{V_2 - 50}{5}$$

$$\text{Constraint equation } V_3 = V_2 + 10i_\phi = 3V_2 - 100$$

$$\Rightarrow V_2 = 60 \text{ v}, \quad V_3 = 80 \text{ v}$$

Chapter-4 Techniques of Circuit Analysis

Example: $V_2 = ?$



We know V_1 and V_3

$$V_1 = 10\text{ v}, \quad V_3 = -20i_\Delta$$

KCL for node-2

$$\frac{V_2}{40} + \frac{V_2 - 10}{10} + \frac{V_2 + 20i_\Delta}{20} = 0$$

KCL for node-1 (constraint equation)

$$-i_\Delta + \frac{10 + 20i_\Delta}{30} + \frac{10 - V_2}{10} = 0$$

$$\Rightarrow -10i_\Delta + 40 - 3V_2 = 0$$

$$\Rightarrow 10i_\Delta = 40 - 3V_2$$

$$\Rightarrow \frac{V_2}{40} + \frac{V_2 - 10}{10} + \frac{V_2 + 80 - 6V_2}{20} = 0$$

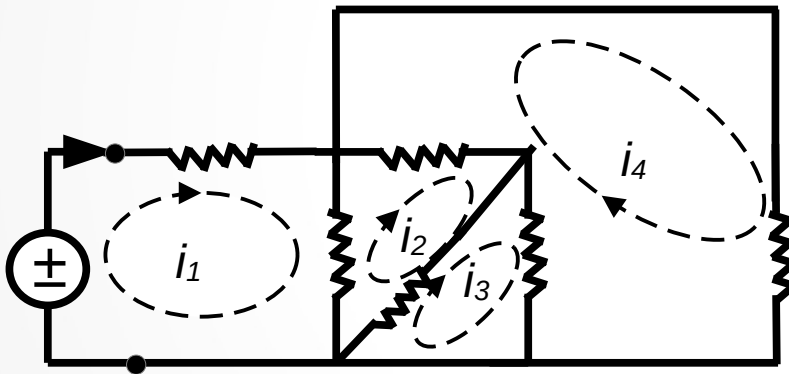
$$\Rightarrow V_2 + 4V_2 - 40 + 2V_2 + 160 - 12V_2 = 0$$

$$\Rightarrow V_2 = \frac{120}{5} = 24\text{ v}$$

Chapter-4 Techniques of Circuit Analysis

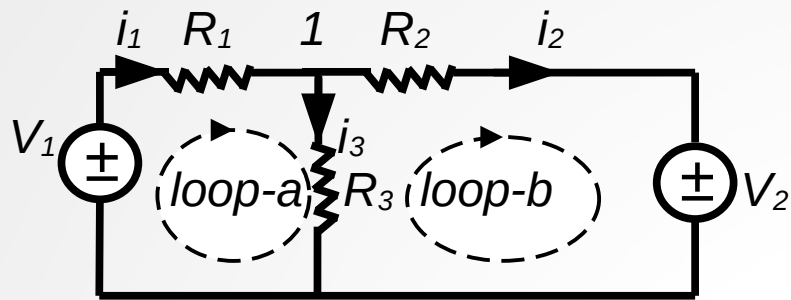
Introduction to the Mesh-Current Method

- This method enable us to describe a circuit in terms of $b_e - (n_e - 1)$ equations.
- A mesh is a loop with no other loops inside it.
- A mesh current is the current that exist only in the perimeter of a mesh.
- The circuit must be planer.



The circuit with mesh currents defined

Chapter-4 Techniques of Circuit Analysis



KCL for node-1

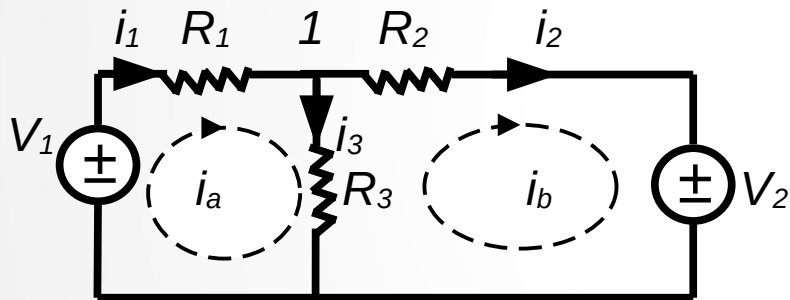
$$i_1 = i_2 + i_3$$

KVL for loop-a

$$-V_1 + i_1 * R_1 + i_3 * R_3 = 0 \Rightarrow V_1 = i_1 * R_1 + i_3 * R_3$$

KVL for loop-b

$$V_2 - i_3 * R_3 + i_2 * R_2 = 0 \Rightarrow -V_2 = i_2 * R_2 - i_3 * R_3$$



Using mesh currents for loop-a

$$-V_1 + R_1 * i_a + R_3 * (i_a - i_b) = 0 \Rightarrow V_1 = R_1 * i_a + R_3 * (i_a - i_b)$$

Using mesh currents for loop-b

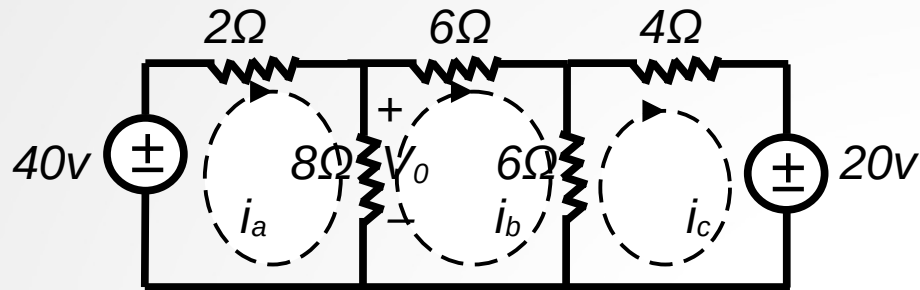
$$V_2 + R_3 * (i_b - i_a) + R_2 * i_b = 0 \Rightarrow -V_2 = R_3 * (i_b - i_a) + R_2 * i_b$$

$$\Rightarrow i_1 = i_a, \quad i_2 = i_b, \quad i_3 = i_a - i_b$$

-Once you know the mesh currents, you also know the branch currents

Chapter-4 Techniques of Circuit Analysis

Example:



- a) Use the mesh-current method to find P_{40v} and P_{20v}
- b) $V_0 = ?$

KVL for Mesh-a

$$-40 + 2i_a + 8(i_a - i_b) = 0$$

KVL for Mesh-b

$$8(i_b - i_a) + 6i_b + 6(i_b - i_c) = 0$$

KVL for Mesh-c

$$6(i_c - i_b) + 4i_c + 20 = 0$$

$$\Rightarrow 10i_a - 8i_b = 40 \Rightarrow i_a = 0.8i_b + 4$$

$$\Rightarrow -8i_a + 20i_b - 6i_c = 0$$

$$\Rightarrow -6i_b + 10i_c = -20 \Rightarrow i_c = 0.6i_b - 2$$

$$\Rightarrow -6.4i_b - 32 + 20i_b - 3.6i_b + 12 = 0$$

$$\Rightarrow 10i_b = 20 \Rightarrow i_b = 2 \text{ A}$$

$$\Rightarrow i_c = 0.6i_b - 2 = -0.8 \text{ A}$$

$$\Rightarrow i_a = 0.8i_b + 4 = 5.6 \text{ A}$$

$$V_0 = 8(i_a - i_b) = 8 * (5.6 - 2) = 28.8 \text{ V}$$

$$P_{40v} = -40 * i_a = -40 * 5.6 = -224 \text{ W (generates)}$$

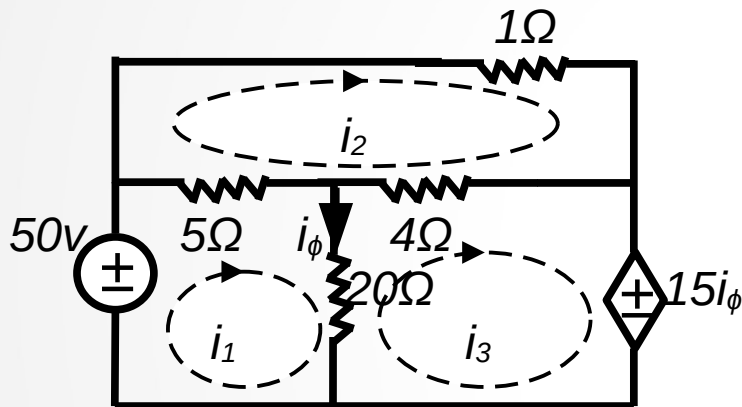
$$P_{20v} = 20 * i_c = 20 * (-0.8) = -16 \text{ W (generates)}$$

Chapter-4 Techniques of Circuit Analysis

Mesh-Current Method and Dependent Sources

-The mesh-current equations must be supplemented by the appropriate constraint equations

Example: $P_{4\Omega} = ?$



Constraint equation: $i_\phi = i_1 - i_3$

KVL for mesh-1

$$-50 + 5(i_1 - i_2) + 20(i_1 - i_3) = 0$$

KVL for mesh-2

$$4(i_2 - i_3) + 5(i_2 - i_1) + i_2 = 0$$

KVL for mesh-3

$$20(i_3 - i_1) + 4(i_3 - i_2) + 15i_\phi = 0$$

$$\Rightarrow 25i_1 - 5i_2 - 20i_3 = 50$$

$$\Rightarrow -5i_1 + 10i_2 - 4i_3 = 0$$

$$\Rightarrow -5i_1 - 4i_2 + 9i_3 = 0$$

$$\Rightarrow 14i_2 = 13i_3$$

$$\Rightarrow -25i_2 + 25i_3 = 50 \Rightarrow i_3 = i_2 + 2$$

$$\Rightarrow i_2 = 26 \text{ A}, \quad i_3 = 28 \text{ A}$$

$$i_1 = 2 * i_2 - 0.8 * i_3 = 29.6 \text{ A}$$

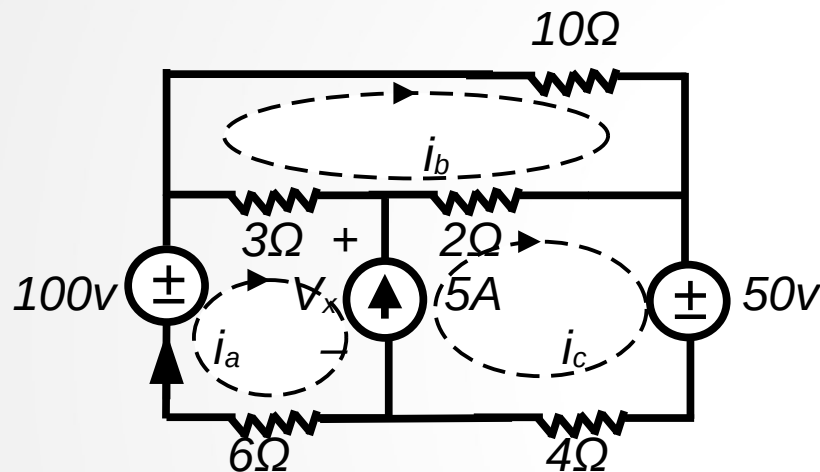
$$P_{4\Omega} = 4 * (i_2 - i_3)^2 = 16 \text{ W}$$

Chapter-4 Techniques of Circuit Analysis

Mesh-Current Method: Some Special Cases

-When a branch includes a current source, the mesh-current method requires some additional manipulations

Example



$$\text{Constraint equation: } i_c - i_a = 5 \text{ A} \Rightarrow i_c = i_a + 5 \text{ A}$$

KVL for mesh a

$$-100 + 3(i_a - i_b) + V_x + 6i_a = 0$$

KVL for mesh c

$$50 + 4i_c - V_x + 2(i_c - i_b) = 0$$

$$\Rightarrow -100 + 3(i_a - i_b) + 6i_a + 50 + 4i_c + 2(i_c - i_b) = 0$$

$$\Rightarrow 9i_a - 5i_b + 6i_c = 50 \quad \text{-----equation-1}$$

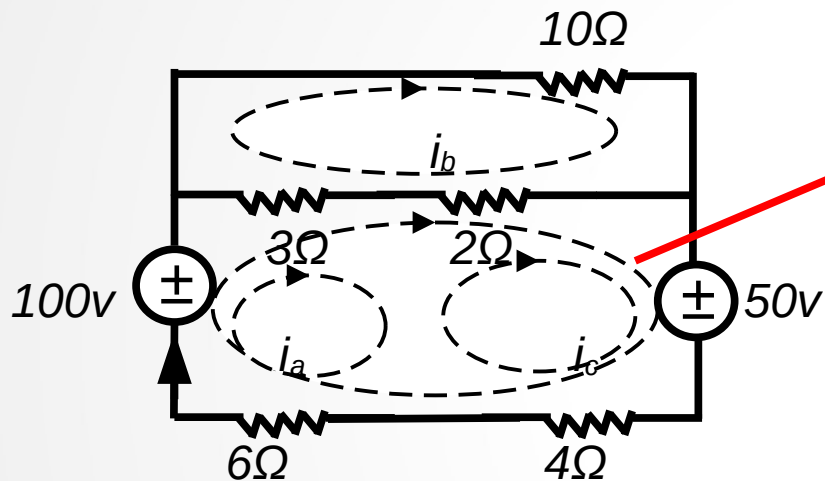
KVL for mesh b

$$3(i_b - i_a) + 10i_b + 2(i_b - i_c) = 0$$

$$\Rightarrow i_a = 1.75 \text{ A}, \quad i_b = 1.25 \text{ A}, \quad i_c = 6.75 \text{ A}$$

Chapter-4 Techniques of Circuit Analysis

The concept of a supermesh



supermesh

We can write equation-1 directly

KVL for the supermesh

$$\begin{aligned} -100 + 3(i_a - i_b) + 2(i_c - i_b) + 50 + 4i_c + 6i_a &= 0 \\ \Rightarrow 9i_a - 5i_b + 6i_c &= 50 \quad \text{-----same as equation-1} \end{aligned}$$

KVL for mesh b

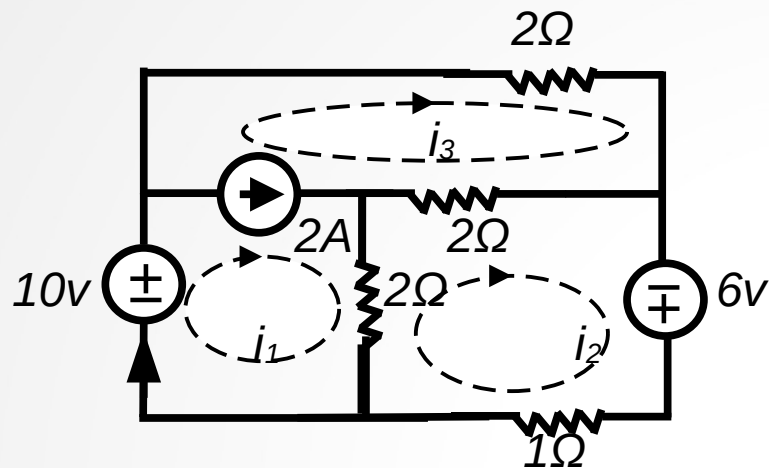
$$3(i_b - i_a) + 10i_b + 2(i_b - i_c) = 0$$

$$\text{Constraint equation: } i_c - i_a = 5 \text{ A} \Rightarrow i_c = i_a + 5 \text{ A}$$

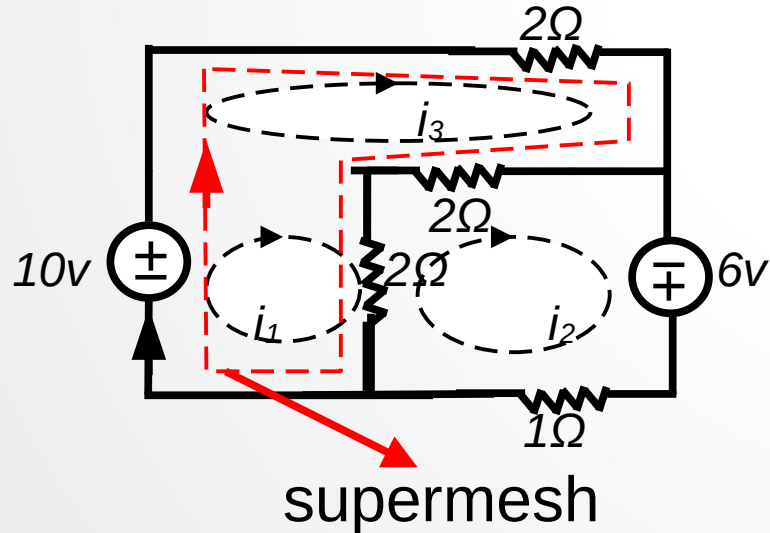
$$\Rightarrow i_a = 1.75 \text{ A}, \quad i_b = 1.25 \text{ A}, \quad i_c = 6.75 \text{ A}$$

Chapter-4 Techniques of Circuit Analysis

Example $P_{1\Omega} = ?$



Use the supermesh concept



Constraint equation: $i_1 - i_3 = 2 \text{ A} \Rightarrow i_1 = i_3 + 2$

KVL for the supermesh

$$-10 + 2i_3 + 2(i_3 - i_2) + 2(i_1 - i_2) = 0$$

$$\Rightarrow 2i_1 - 4i_2 + 4i_3 = 10$$

$$\Rightarrow -4i_2 + 6i_3 = 6$$

KVL for mesh 2

$$-6 + i_2 + 2(i_2 - i_1) + 2(i_2 - i_3) = 0$$

$$\Rightarrow -2i_1 + 5i_2 - 2i_3 = 6$$

$$\Rightarrow 5i_2 - 4i_3 = 10$$

$$\Rightarrow 7i_3 = 35 \Rightarrow i_3 = 5 \text{ A}$$

$$\Rightarrow i_2 = 6 \text{ A}, \quad i_1 = 7 \text{ A}$$

$$P_{1\Omega} = i_2^2 * 1 = 6^2 = 36 \text{ W}$$

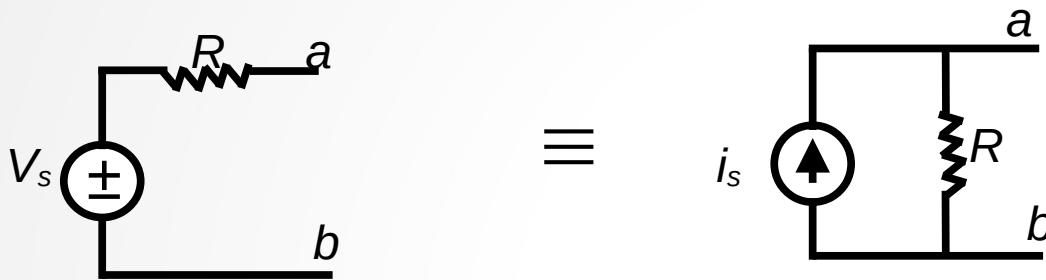
The Node-Voltage Method versus the Mesh-Current Method

- Check the number of equations
- Does the circuit contain supernode?
- Does the circuit contain supermesh?

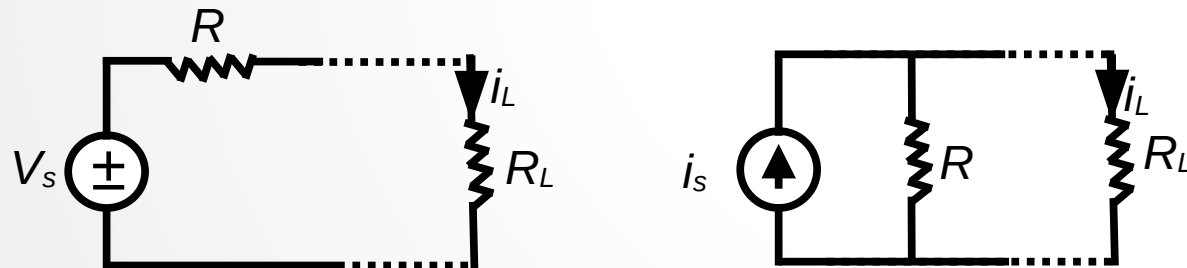
Chapter-4 Techniques of Circuit Analysis

Source Transformation

-A source transformation allows a voltage source in series with a resistor to be replaced by a current source in parallel with the same resistor or vice versa.



Assume R_L is connected between nodes a and b



i_L must be the same for both circuit

$$\Rightarrow i_L = \frac{V_s}{R + R_L}, \quad i_L = \frac{R}{R + R_L} i_s$$

$$\Rightarrow \frac{V_s}{R + R_L} = \frac{R}{R + R_L} i_s$$

$$\Rightarrow i_s = \frac{V_s}{R} \quad \text{or}$$

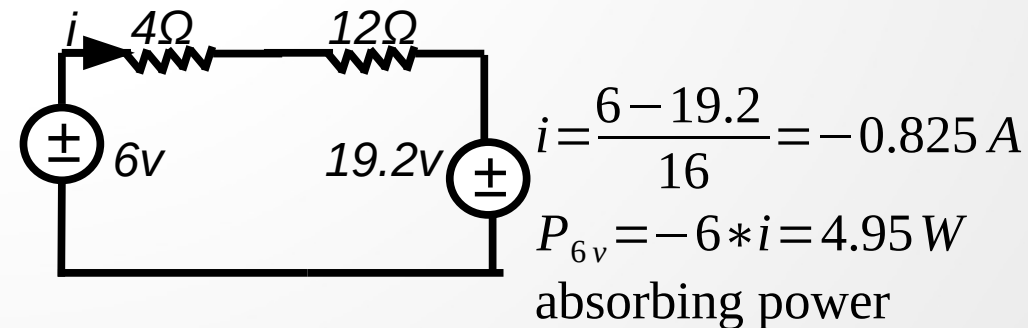
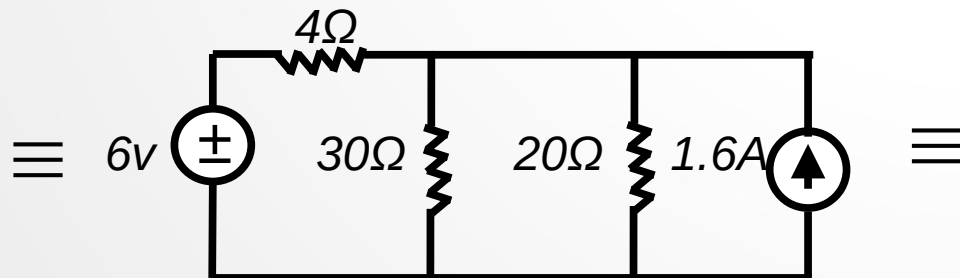
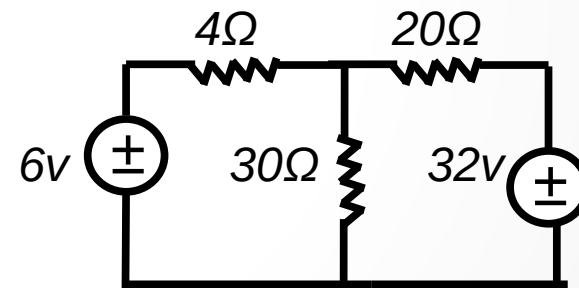
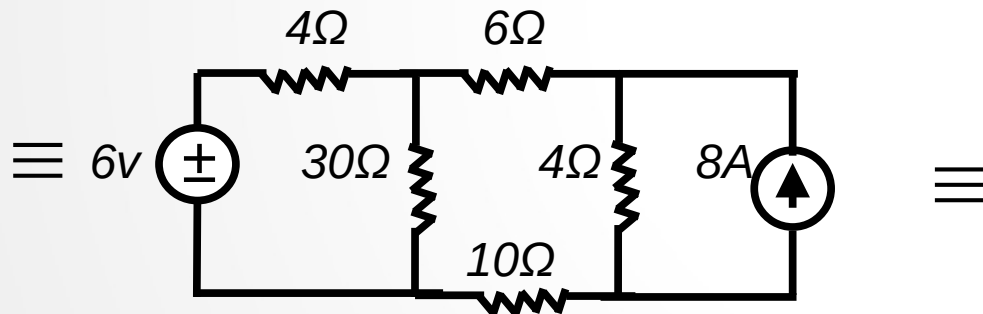
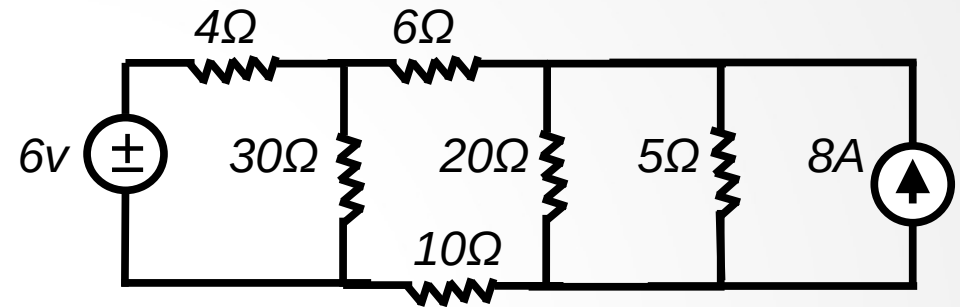
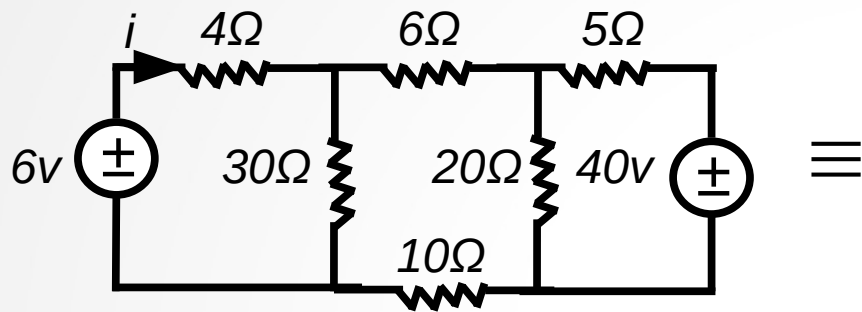
$$V_s = i_s * R$$

Chapter-4 Techniques of Circuit Analysis

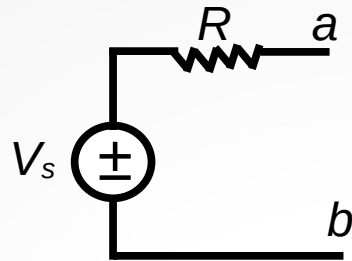
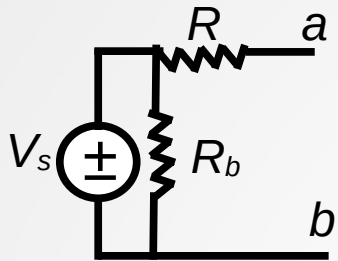
Example

a) power associated with the 6v voltage source=?

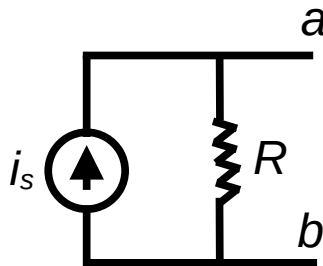
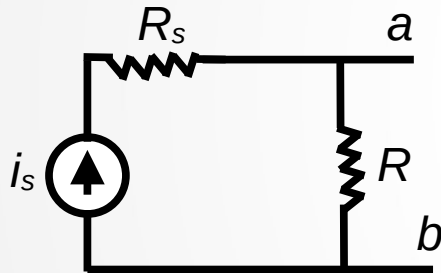
b) Is the 6v voltage source absorbing or delivering power?



Chapter-4 Techniques of Circuit Analysis



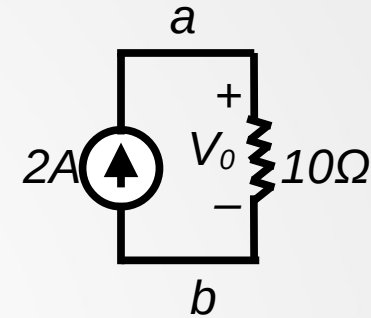
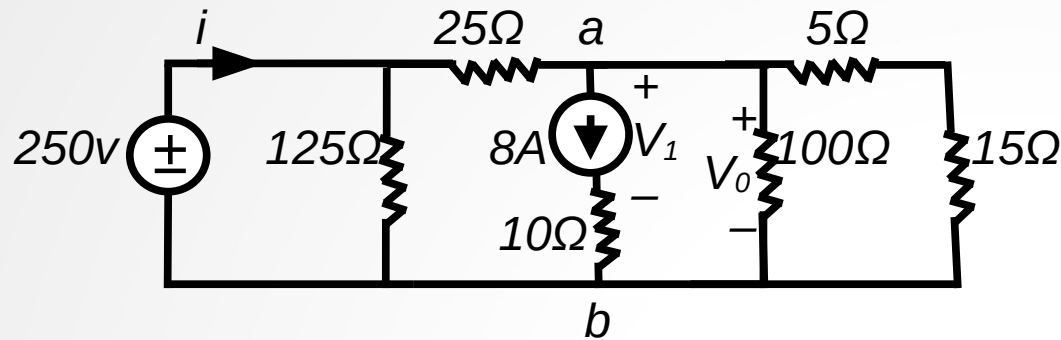
Equivalent with respect to terminals a and b .



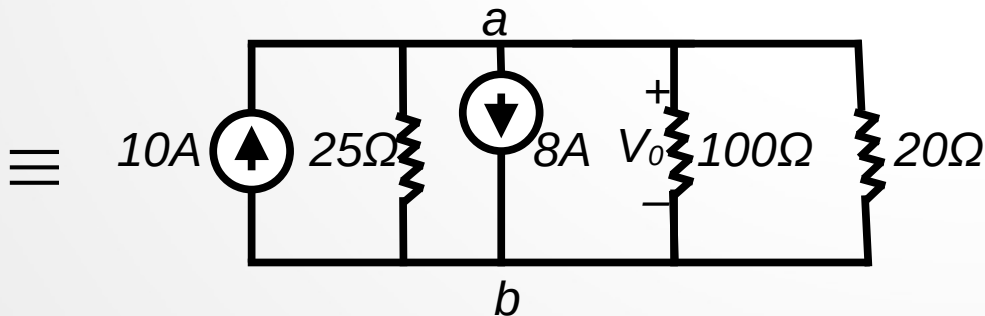
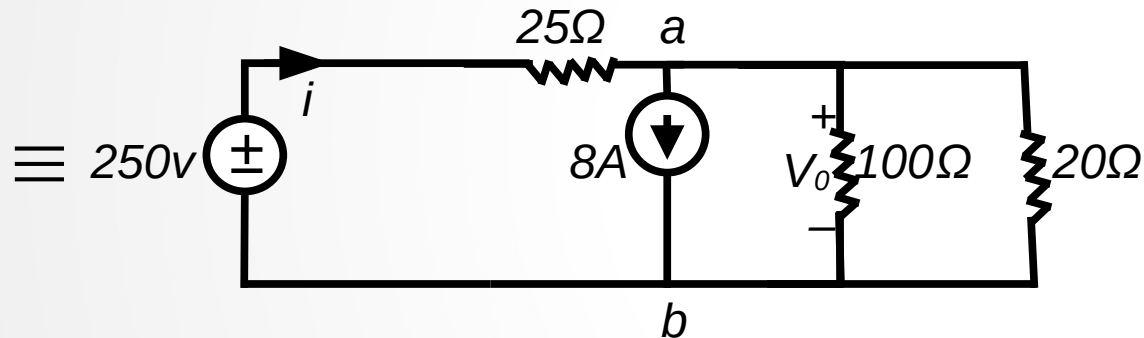
Equivalent with respect to terminals a and b .

Chapter-4 Techniques of Circuit Analysis

Example: $V_0 = ?$, $P_{250V} = ?$, $P_{8A} = ?$



Remove 125Ω and 10Ω



$$\Rightarrow V_0 = 20 \text{ V}$$

$$i = \frac{250}{125} + \frac{250 - V_0}{25} = 2 + \frac{230}{25} = 11.2 \text{ A}$$

$$P_{250V} = -250 * i = -2800 \text{ W}$$

(Delivering power)

$$V_1 = ? \quad -V_1 + V_0 - 10 * 8 = 0$$

$$\Rightarrow V_1 = V_0 - 80 = 20 - 80 = -60 \text{ V}$$

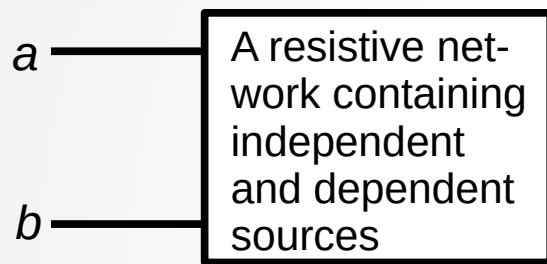
$$P_{8A} = V_1 * 8 = -480 \text{ W}$$

(Delivering power.)

Chapter-4 Techniques of Circuit Analysis

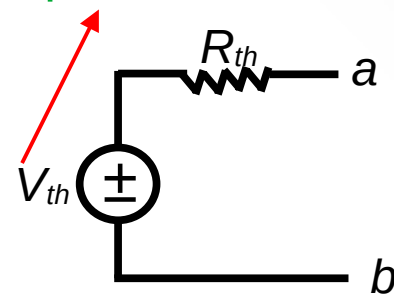
Thevenin and Norton Equivalents

- We want to concentrate on what happens at a specific pair of terminals
- Thevenin and Norton equivalents are circuit simplification techniques that focus on terminal behavior.



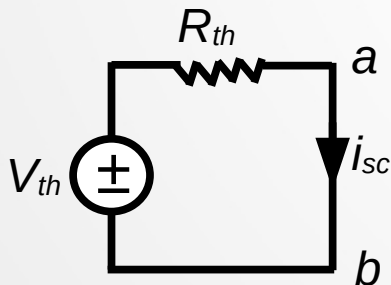
$\equiv$

Open circuit voltage



Thevenin equivalents circuit

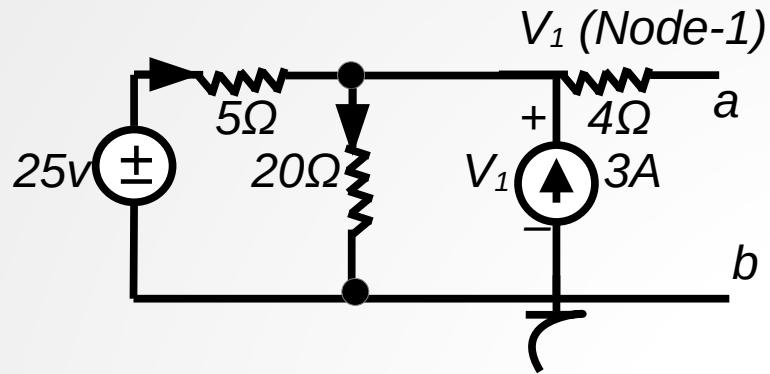
- If we connect the same load across the terminal a,b of each circuit, we get the same voltage and current at the terminals of the load.



$$i_{sc} = \frac{V_{th}}{R_{th}} \Rightarrow R_{th} = \frac{V_{th}}{i_{sc}}$$

Chapter-4 Techniques of Circuit Analysis

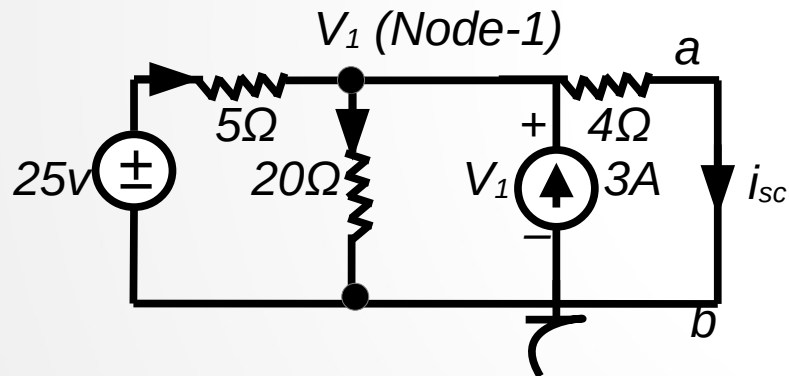
Example: Finding a Thevenin Equivalent



KVL for Node-1

$$\frac{V_1 - 25}{5} + \frac{V_1}{20} - 3 = 0 \Rightarrow V_1 = 32 \text{ v}$$

$$V_{th} = V_{ab} = V_1 = 32 \text{ v}$$

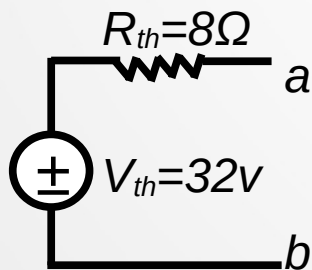


KVL for Node-1

$$\frac{V_1 - 25}{5} + \frac{V_1}{20} + \frac{V_1}{4} - 3 = 0 \Rightarrow V_1 = 16 \text{ v}$$

$$i_{sc} = \frac{V_1}{4} = \frac{16}{4} = 4 \text{ A}$$

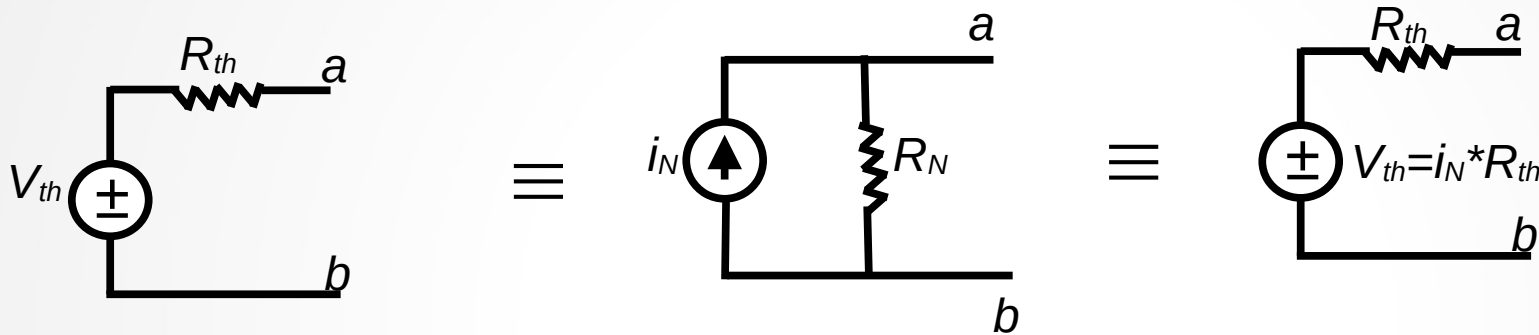
$$R_{th} = \frac{V_{th}}{i_{sc}} = \frac{32}{4} = 8 \Omega$$



Chapter-4 Techniques of Circuit Analysis

The Norton Equivalent

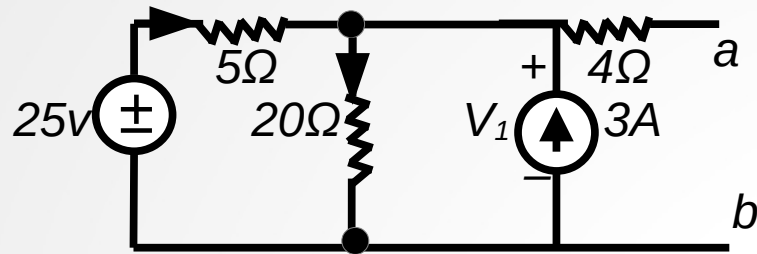
-The Norton equivalent circuit consists of an independent current source in parallel with the Norton equivalent resistance R_N .



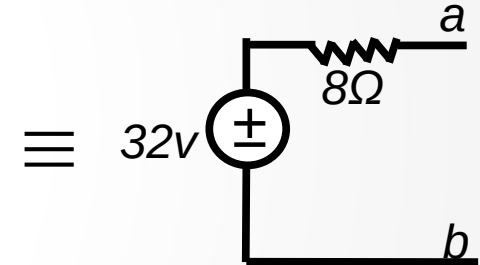
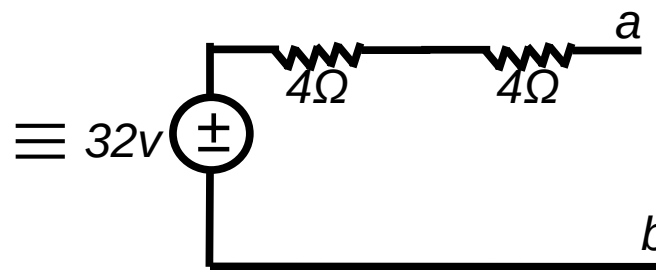
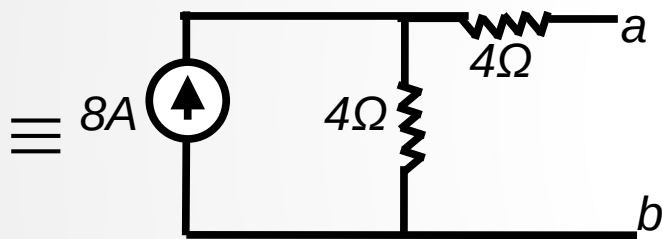
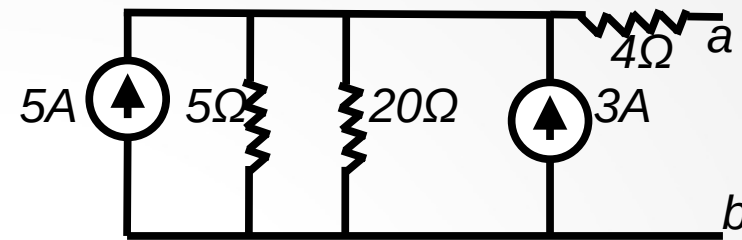
$$R_N = R_{th}, \quad i_N = i_{sc} \text{ (short circuit current)}$$

Chapter-4 Techniques of Circuit Analysis

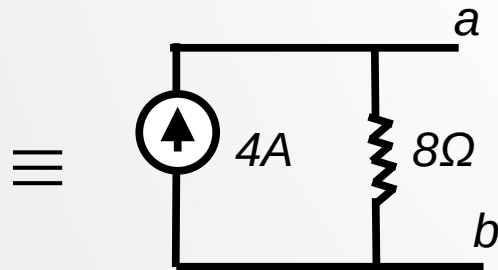
Example: (Using source transformation)



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Thevenin Equivalent

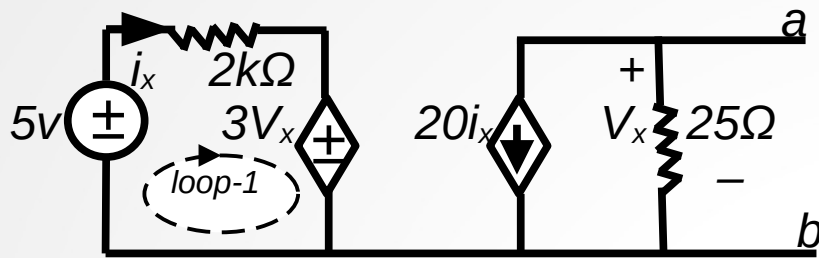


Norton Equivalent

$$\Rightarrow R_N = R_{th} = 8\Omega, \quad i_N = i_{sc} = 4A, \quad V_{th} = 32V$$

Chapter-4 Techniques of Circuit Analysis

Example: Find the Thevenin equivalent



$$V_{th} = V_{ab} = V_x = -20i_x * 25 = -500i_x$$

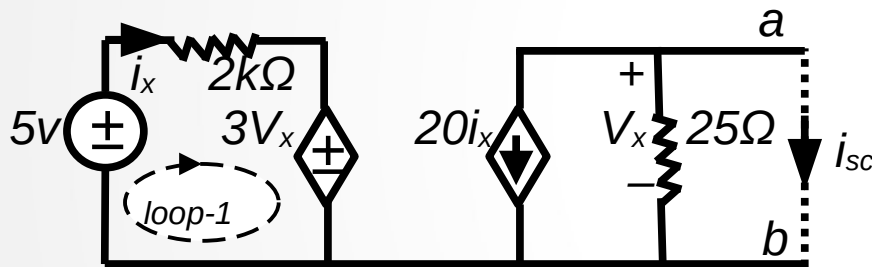
KVL for loop-1

$$-5 + 2000i_x + 3V_x = 0 \Rightarrow 2000i_x - 1500i_x = 5$$

$$\Rightarrow i_x = 10 \text{ mA}$$

$$V_{th} = V_{ab} = V_x = -500i_x = -5 \text{ V}$$

$i_{sc} = ?$ (short circuit current)



$$V_{ab} = V_x = 0$$

KVL for loop-1

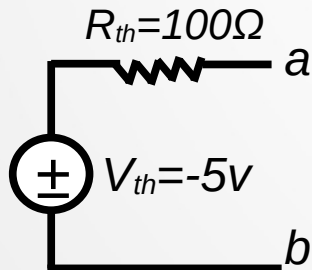
$$-5 + 2000i_x + 3V_x = 0 \Rightarrow i_x = \frac{5}{2000}$$

KCL for node a

$$20i_x + i_{sc} = 0$$

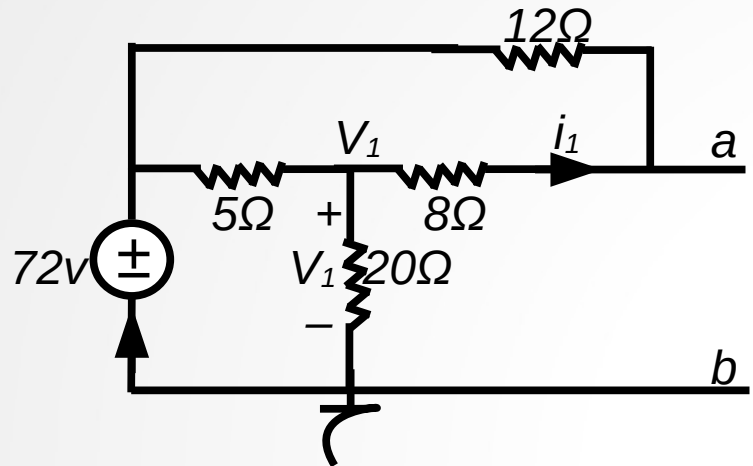
$$i_{sc} = -20 * \frac{5}{2000} = -50 \text{ mA} = -0.05 \text{ A}$$

$$R_{th} = \frac{V_{th}}{i_{sc}} = \frac{-5}{-0.05} = 100 \Omega$$



Chapter-4 Techniques of Circuit Analysis

Example: Find the Thevenin equivalent circuit with respect to the terminals a and b.



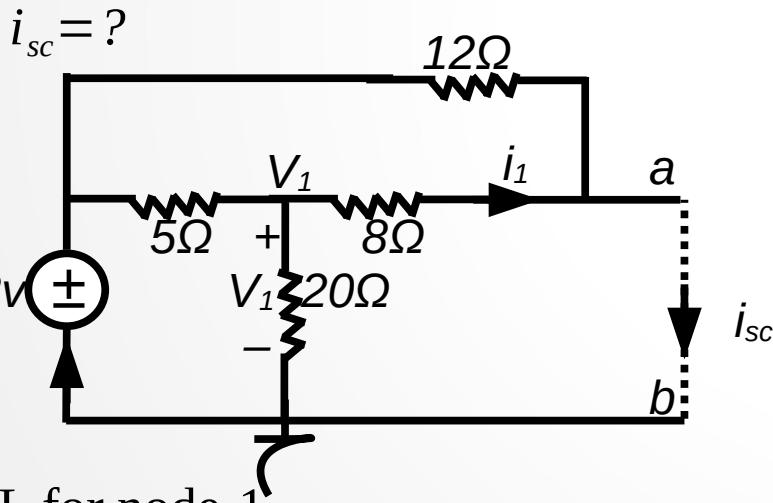
KCL for node-1

$$\frac{V_1}{20} + \frac{V_1 - 72}{5} + \frac{V_1 - 72}{8 + 12} = 0$$

$$\Rightarrow V_1 + 4V_1 - 288 + V_1 - 72 = 0 \Rightarrow V_1 = \frac{360}{60} = 60 \text{ v}$$

$$i_1 = \frac{V_1 - 72}{20} = -0.6 \text{ A}$$

$$V_{ab} = V_{th} = V_1 - 8i_1 = 60 + 8 \times 0.6 = 64.8 \text{ v}$$



KCL for node-1

$$\frac{V_1}{20} + \frac{V_1 - 72}{5} + \frac{V_1}{8} = 0 \Rightarrow 2V_1 + 8V_1 - 576 + 5V_1 = 0$$

$$\Rightarrow 2V_1 + 8V_1 - 576 + 5V_1 = 0$$

$$\Rightarrow V_1 = \frac{576}{15} = 38.4 \text{ v}$$

KCL for node a

$$i_{sc} = \frac{V_1}{8} + \frac{72}{12} = \frac{144 + 115.2}{24} = 10.8 \text{ A}$$

$$R_{th} = \frac{V_{th}}{i_{sc}} = \frac{64.8}{10.8} = 6 \Omega$$

Chapter-4 Techniques of Circuit Analysis

More on Deriving a Thevenin Equivalent to find R_{th}

-If the circuit contains only dependent sources, we first deactivate all independent sources and calculate the equivalent resistance. (Replace each independent voltage source by a short circuit. Replace each independent current source by an open circuit.)

-If the circuit contains dependent sources, we first deactivate all independent sources and then we apply either a test voltage source or a test current source to the Thevenin terminals a and b .

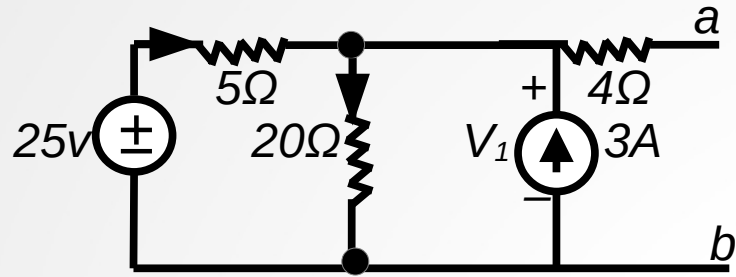
$$\Rightarrow R_{th} = \frac{V_{ab}}{i_{ab}}$$

V_{ab} is the voltage rise from terminal b to terminal a

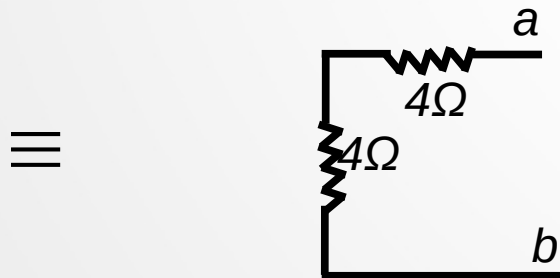
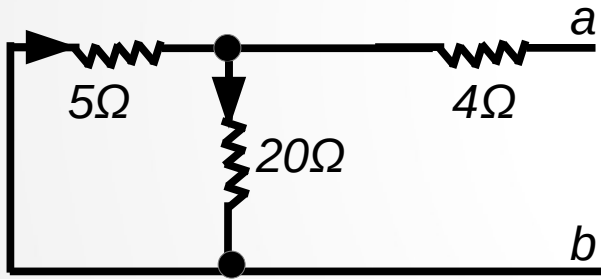
i_{ab} is the current flow from terminal b to terminal a

Chapter-4 Techniques of Circuit Analysis

Example: $R_{th} = ?$



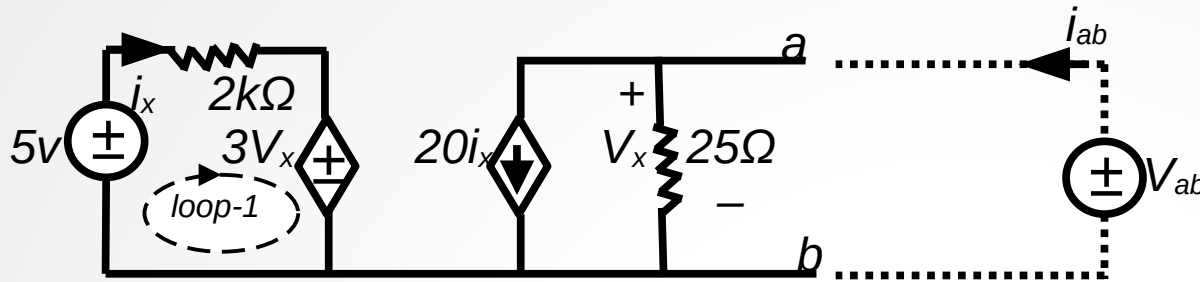
Deactivate all independent sources



$$\Rightarrow R_{th} = R_{ab} = R_N = 8\Omega$$

Chapter-4 Techniques of Circuit Analysis

Example: (dependent sources) $R_{th}=?$ (Seen from terminals a and b)



$$\Rightarrow R_{th} = \frac{V_{ab}}{i_{ab}}$$

KVL for loop-1

$$2000 * i_x + 3V_x = 0$$

$$\Rightarrow i_x = -3 \frac{V_x}{2000} = -3 \frac{V_{ab}}{2000}$$

KCL for node-a

$$i_{ab} = 20i_x + \frac{V_{ab}}{25}$$

$V_{ab} = V_x$ because V_{ab} and 25Ω are connected in parallel

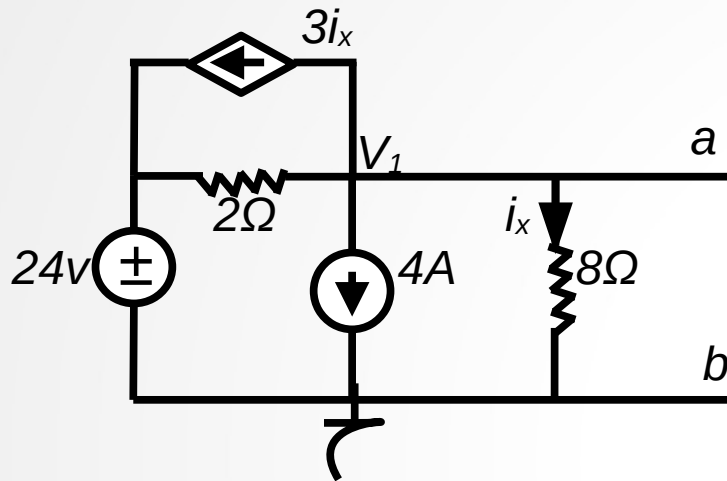
$$\Rightarrow i_{ab} = \frac{-60}{2000} V_{ab} + \frac{V_{ab}}{25}$$

$$\Rightarrow i_{ab} = \frac{-3V_{ab}}{100} + \frac{4V_{ab}}{100} = \frac{V_{ab}}{100}$$

$$\Rightarrow R_{th} = \frac{V_{ab}}{i_{ab}} = 100\Omega$$

Chapter-4 Techniques of Circuit Analysis

Example: $R_{th}=?$, $V_{th}=?$ (seen from terminals a and b)



Constraint equation: $i_x = \frac{V_1}{8} \Rightarrow V_1 = 8i_x$

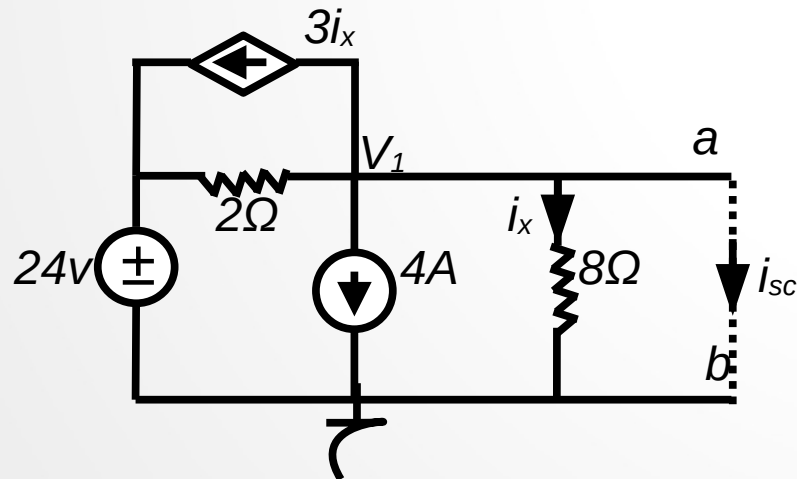
KCL for node-1 $\frac{V_1 - 24}{2} + 4 + i_x + 3i_x = 0$

$\Rightarrow \frac{8i_x - 24}{2} + 4 + i_x + 3i_x = 0 \Rightarrow 8i_x - 8 = 0 \Rightarrow i_x = 1A$

$V_{ab} = V_{th} = V_1 = 8i_x = 8V$

$R_{th} = ?$

Method-1:



$V_1 = 0V$

Constraint equation: $i_x = \frac{V_1}{8} \Rightarrow i_x = 0A$

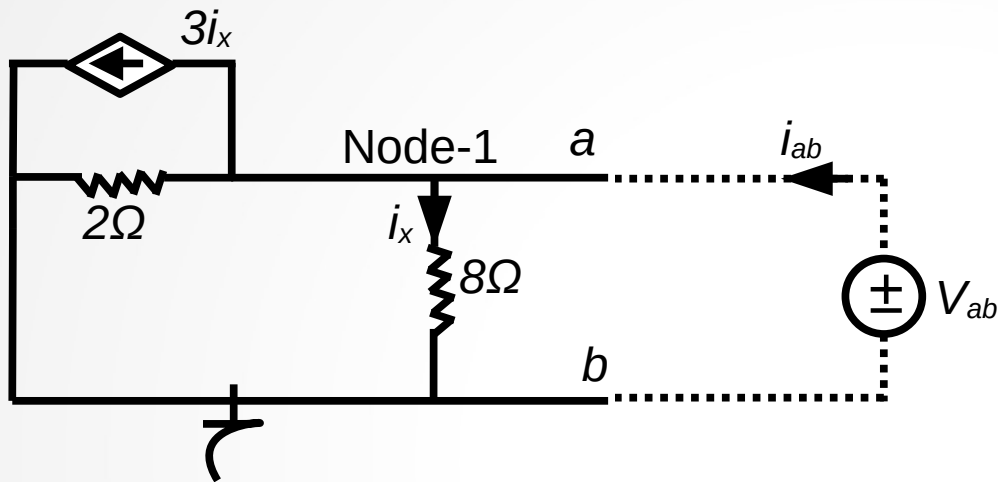
KCL for node-1

$\frac{V_1 - 24}{2} + 4 + i_x + 3i_x + i_{sc} = 0 \Rightarrow i_{sc} = 8A$

$\Rightarrow R_{th} = \frac{V_{th}}{i_{sc}} = \frac{8}{8} = 1\Omega$

Chapter-4 Techniques of Circuit Analysis

Method-2:



$$\text{Constraint equation: } i_x = \frac{V_{ab}}{8}$$

KCL for node-1

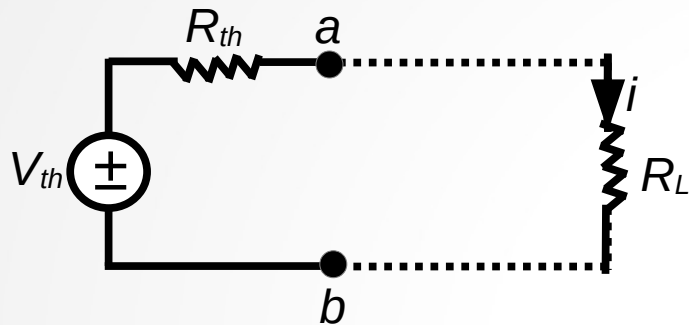
$$\frac{V_{ab}}{2} + \frac{V_{ab}}{8} + 3i_x - i_{ab} = 0$$

$$\Rightarrow \frac{V_{ab}}{2} + \frac{V_{ab}}{8} + 3\frac{V_{ab}}{8} = i_{ab} \Rightarrow R_{th} = \frac{V_{ab}}{i_{ab}} = 1\Omega$$

Chapter-4 Techniques of Circuit Analysis

Maximum Power Transfer

-Find the value of R_L that allow us to transfer maximum power to the R_L



P_{max} = maximum power for the R_L

$$\Rightarrow P_{max} = i^2 * R_L = \left(\frac{V_{th}}{R_{th} + R_L} \right)^2 * R_L$$
$$= \left(\frac{V_{th}}{2 R_L} \right)^2 * R_L = \frac{V_{th}^2}{4 R_L}$$

$$P_{R_L} = i^2 * R_L = \left(\frac{V_{th}}{R_{th} + R_L} \right)^2 * R_L$$

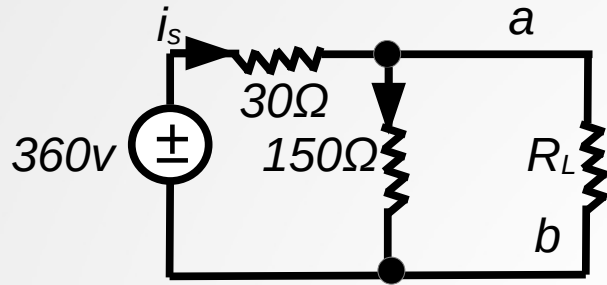
$$\frac{dP_{R_L}}{dR_L} = V_{th}^2 \frac{(R_{th} + R_L)^2 - 2 R_L * (R_{th} + R_L)}{(R_{th} + R_L)^2}$$

$$\frac{dP_{R_L}}{dR_L} = 0 \Rightarrow (R_{th} + R_L)^2 = 2 R_L * (R_{th} + R_L)$$

$$\Rightarrow R_L = R_{th} \text{ for maximum power transfer}$$

Chapter-4 Techniques of Circuit Analysis

Example

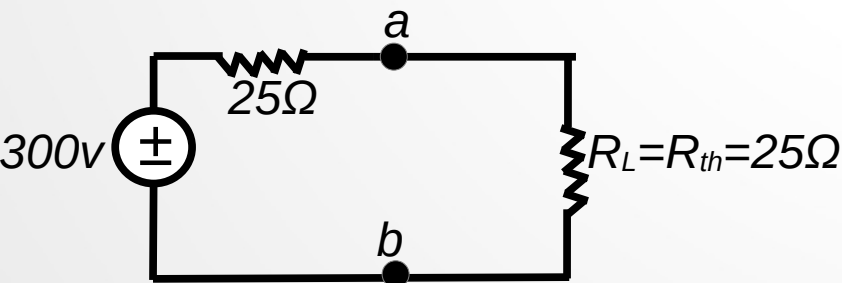


- a) Find the value of R_L that results in maximum power being delivered to R_L
- b) $P_{\max}=?$
- c) What percentage of the power delivered by the 360V voltage source reaches R_L

Solution: First, let's find the thevenin equivalent seen from terminals a and b.

$$V_{th} = \frac{150}{150 + 30} * 360 = 300 \text{ V}$$

$$R_{th} = \frac{30 * 150}{30 + 150} = 25 \Omega$$



$$(b) \quad P_{\max} = P_{R_L} = \frac{V_{th}^2}{4 R_L} = \frac{300^2}{4 * 25} = 900 \text{ W}$$

$$(c) \quad V_{ab} = \frac{300 * 25}{25 + 25} = 150 \text{ V}$$

$$i_s = \frac{V_{ab}}{150} + \frac{V_{ab}}{25} = 7 \text{ A}$$

$$\Rightarrow P_{360} = -360 * i_s = -2520 \text{ W} \text{ generates power}$$

The percentage of the source power delivered

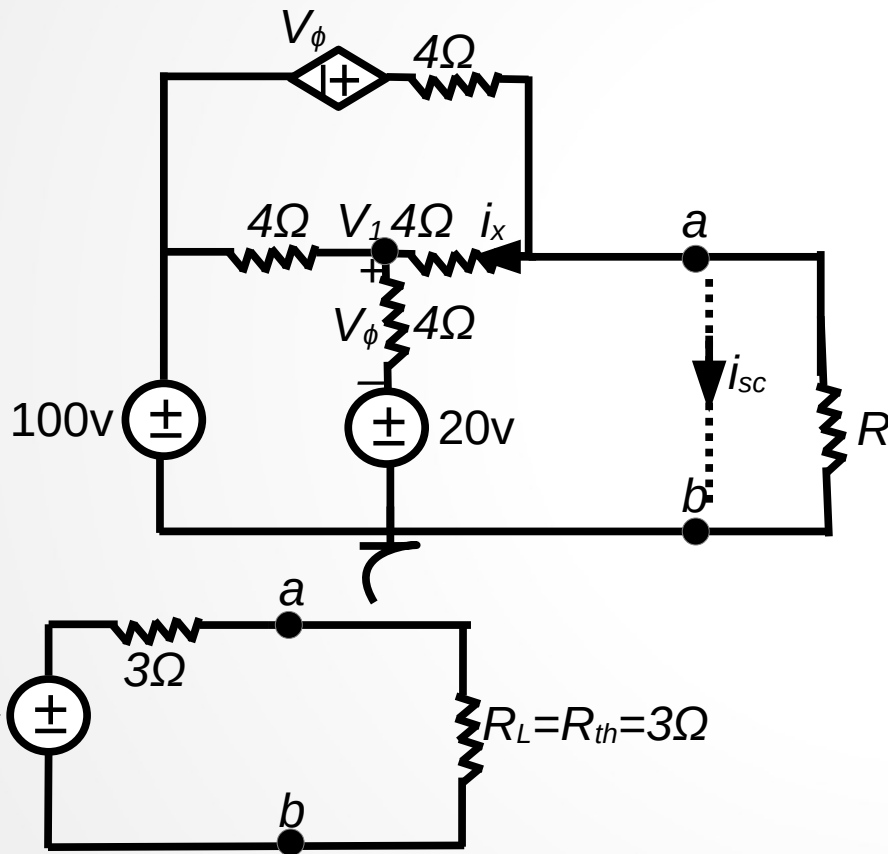
$$\text{to the load is: } \frac{900}{2520} * 100 = 35.71 \%$$

Chapter-4 Techniques of Circuit Analysis

Example

a) Find the value of R that enables the circuit to deliver maximum power to the R

b) $P_{max}=?$



$$R_{th} = \frac{V_{th}}{i_{sc}} = \frac{120}{40} = 3\Omega = R \quad P_{max} = \frac{V_{th}^2}{4R} = \frac{120^2}{12} = 1200 \text{ W}$$

Constraint equation: $V_1 = 20 + V_\phi \Rightarrow V_\phi = V_1 - 20$

KCL for node-1

$$\frac{V_1 - 100}{4} + \frac{V_1 - 20}{4} + \frac{V_1 - (V_\phi + 100)}{8} = 0$$

$$2V_1 - 200 + 2V_1 - 40 + V_1 - V_1 + 20 - 100 = 0$$

$$\Rightarrow 4V_1 = 320 \Rightarrow V_1 = 80 \text{ V}$$

$$i_x = \frac{100 + V_\phi - V_1}{8} = \frac{80}{8} = 10 \text{ A}$$

$$V_{th} = V_{ab} = V_1 + 4 * i_x = 80 + 4 * 10 = 120 \text{ V}$$

$i_{sc} = ?$

$$\text{KCL for node-1} \quad \frac{V_1 - 20}{4} + \frac{V_1}{4} + \frac{V_1 - 100}{4} = 0$$

$$\Rightarrow V_1 = 40 \text{ V} \Rightarrow V_\phi = 20 \text{ V}$$

$$\text{KCL for node-a} \quad i_{sc} = \frac{V_1}{4} + \frac{100 + V_\phi}{4} = 40 \text{ A}$$